RAJALAKSHMI ENGINEERING COLLEGE

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|---|---|
| **Started on** | Friday, 7 August 2026, 9:36 AM |
| **State** | Finished |
| **Completed on** | Friday, 21 August 2026, 8:35 AM |
| **Time taken** | 13 days 22 hours |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include<stdio.h>
void function(int n)
{
    int c= 0;
    int count=1;
    count++;
    for(int i=n/2; i<n; i++){
        count+=2;
        for(int j=1; j<n; j = 2 * j){
               count+=2;
            for(int k=1; k<n; k = k * 2){
                c++;
                count+=2;
            }
       }
    }
    printf("%d",count);
}
int main(){
    int n;
    scanf("%d",&n);
    function(n);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4 | 30 | 30 | ✔ |
| ✔ | 10 | 212 | 212 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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